

nieuwe rij (bij selectie)			nieuwe rij ('alternatief')		randen tekenen		links >> distributeur		update prijzen (API)				update status 'skip'			
Onder deel	Type	Beschrijving	Component	Opmerking	Tr Waarde	Tr Vorm	Distributeu r	#	📦 € (1)	📦 €€	📦 Status	referenc e	Symbool	Footprint		
STM32	IC	Microcontroller	STM32L431CCT6TR	Cortex-M4 256KB flash 64KB ram LQFP-48 Min. 5	-	LQFP48 (SOT313-2)	Farnell	5	€ 4,38	€ 0,00	skip	U3	STM32L431CCT	Package_QFP:LQFP-48_7x7mm_P0.5mm		
STM32	IC, alternatief	Microcontroller	STM32L431CCT6TR	Cortex-M4 256KB flash 64KB ram LQFP-48 Min. 5	-	LQFP48 (SOT313-2)	Digikey	5	€ 4,82	€ 24,10	correct, in kicad, niet op voorraad	U3	STM32L431CCT	Package_QFP:LQFP-48_7x7mm_P0.5mm		
STM32	IC, alternatief	Microcontroller	STM32L431CCT6TR	Cortex-M4 256KB flash 64KB ram LQFP-48 Min. 5	-	LQFP48 (SOT313-2)	Mouser	5	€ 5,02	€ 0,00	skip	U3	STM32L431CCT	Package_QFP:LQFP-48_7x7mm_P0.5mm		
STM32	C, SMD	Ontkoppel (VDD)	CC0603KRX7R6BB104	Zie datasheet 6.1.6 Power supply scheme (p.85)	100nF	0603	Mouser	3	€ 0,12	€ 0,00	correct, in kicad, skip	C14, C15, C16	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD, alternatief	Ontkoppel (VDD)	CC0603KRX7R6BB104	Zie datasheet 6.1.6 Power supply scheme (p.85)	100nF	0603	Mouser	3	€ 0,12	€ 0,36	correct, in kicad	C14, C15, C16	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD	Ontkoppel (VDD)	CC0603KPX5R6BB475	idem	4,7µF	0603	Mouser	1	€ 0,24	€ 0,24	correct, in kicad	C41	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD	Ontkoppel (VDDA)	CC0603KPX5R6BB105	idem	1µF	0603	Mouser	1	€ 0,11	€ 0,11	correct, in kicad	C17	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD	Ontkoppel (VDDA)	CC0603KRX7R6BB103	idem	10nF	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	C18	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD	Ontkoppel (VDDA)	CC0603KRX7R6BB104	Voor Vref, maar is intern gelinkt aan VDDA	100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C40	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD, alternatief	Ontkoppel (VDDA)	CC0603KRX7R6BB104	Voor Vref, maar is intern gelinkt aan VDDA	100nF	0603	Mouser	1	€ 0,12	€ 0,12	correct, in kicad	C40	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD	Anti-aliasing VBAT_SENSE	CC0603KRX7R6BB103	Anti-aliasing filter op ADC-ingang PA0	10nF	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	C28	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD	Reset circuit	CC0603KRX7R6BB104	Zie datasheet 6.3.15 NRST pin characteristics (p.136)	100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C19	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	C, SMD, alternatief	Reset circuit	CC0603KRX7R6BB104	Zie datasheet 6.3.15 NRST pin characteristics (p.136)	100nF	0603	Mouser	1	€ 0,12	€ 0,12	correct, in kicad	C19	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
STM32	R, SMD	Batterijspanning deler (hoog)	RC0603FR-07220KL	VBAT_SENSE-deler: 220K/(220K+100K) × Vbat → MCU PA0 (ADC1_IN5)	220K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R18	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
STM32	R, SMD	Batterijspanning deler (laag)	RC0603FR-07100KL	Zie hierboven	100K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R19	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
STM32	R, SMD	I²C pullup	RC0603FR-074K7L	Accelerometer module bevat al interne pullups — alleen plaatsen bij overschakelen naar chip Toch toevoegen aan schema en pads voorzien??	4,7K	0603	Mouser	1	€ 0,10	€ 0,00	skip		Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
STM32	R, SMD	I²C pullup	RC0603FR-074K7L	Accelerometer module bevat al interne pullups — alleen plaatsen bij overschakelen naar chip Toch toevoegen aan schema en pads voorzien??	4,7K	0603	Mouser	1	€ 0,10	€ 0,00	skip		Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
STM32	R, SMD	Bootknop pulldown	RC0603FR-0710KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,00	correct, skip	R17	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
STM32	R, SMD, alternatief	Bootknop pulldown	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R17	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
STM32	L, SMD	Ferrite bead VDDA (FB1)	BLM18KG601SN1D	600 Ω @ 100 MHz; filtert VDDA-pin MCU af van +3V3-rail	600Ω@100M Hz	0603	Digikey	1	€ 0,09	€ 0,09	correct, in kicad	FB1	Device:Ferrite_Be	Inductor_SMD:L_0603_1608Metric Inductor_SMD:L_0603_1608Metric_Pad1.05x0.95mm_HandSol		
STM32	Knop, THT	Reset knop	TS02-66-43-BK-160-LCR-D	-	-	SPST momentary tactile, 6×6 mm THT	Mouser	1	€ 0,09	€ 0,09	in bezit, in kicad	SW1	Switch:SW_Push	s		
STM32	Knop, THT	Boot knop	TS02-66-43-BK-160-LCR-D	-	-	SPST momentary tactile, 6×6 mm THT	Mouser	1	€ 0,09	€ 0,09	in bezit, in kicad	SW2	Switch:SW_Push	Button_Switch_THT:SW_PUSH_6mm		
LoRa	Module, SMD	LoRa + BLE module MHF4-connector	WisDuo RAK11720 (MHF4 connector)	LoRa + BLE, MHF4 connector		half hole PCB breakout (1mm)	Digikey	1	€ 12,76	€ 12,76	correct, in kicad	U9	https://www.snar	https://www.snapeda.com/parts/RAK11720/RAK/view-part/?w		
LoRa	C, SMD	Ontkoppel (VDD)	CC0603KRX5R6BB106	-	10µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad, niet op voorraad	C37	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
LoRa	C, SMD	Ontkoppel (VDD)	CC0603KPX5R6BB105	-	1µF	0603	Mouser	1	€ 0,11	€ 0,11	correct, in kicad	C38	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
LoRa	C, SMD	Ontkoppel (VDD)	CC0603KRX7R6BB104	-	100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C39	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
LoRa	C, SMD, alternatief	Ontkoppel (VDD)	CC0603KRX7R6BB104	-	100nF	0603	Mouser	1	€ 0,12	€ 0,10	correct, in kicad	C39	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSol		
LoRa	R, SMD	SWO pulldown	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R25	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
LoRa	R, SMD	#RST Pullup	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R24	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSol		
LoRa antenne	Antenne	FPC antenne	FPC antenne	2.2dBi, VSWR 1.5 Optie voor 5, 10 of 15cm kabel		FPC antenne (5, 10 of 15cm kabel)	Farnell	1		€ 0,00	correct		-	-		

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Onder deel	Type	Beschrijving	Component	Opmerking	T _r Waarde	T _r Vorm	Distributeu r	#	📄 € (1)	📄 €€	📄 Status	referenc e	Symbool	Footprint		
BLE antenne	Antenne	FPC antenne	AANI-FB-0086-0100W (100mm)	Optie voor 10, 15 of 20cm kabel		FPC antenne (10, 15 of 20cm kabel)	Digikey	1	€ 2,57	€ 2,57	correct		-	-		
EEC chip	IC	EEC chip	ATECC608B-SSHDA-T	-		8-SOIC	Digikey	1	€ 0,79	€ 0,79	correct, in kicad	U5	ATECC608B-SSHD	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm		
EEC chip	IC, alternatief	EEC chip	ATECC608B-SSHDA-T	-		8-SOIC	Mouser	1	€ 0,77	€ 0,00	skip, correct	U5	ATECC608B-SSHD	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm		
EEC chip	IC, alternatief	EEC chip	ATECC608B-SSHDA-T	-		8-SOIC	Farnell	1		€ 0,00	skip, correct	U5	ATECC608B-SSHD	Package_SO:SOIC-8_3.9x4.9mm_P1.27mm		
EEC chip	C, SMD	Ontkoppel	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C27	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
EEC chip	C, SMD, alternatief	Ontkoppel	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,10	correct, in kicad	C27	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
EEC chip	Breakout, THT	SOIC8-to-DIP8 adapter	FIT0290	inclusief pin headers	-	DIP-8	Digikey	1	€ 0,48	€ 0,48	correct, in kicad	-	-	-		
EEC chip	IC socket, THT	DIP8 socket	1-2199298-2		-	DIP-8	Mouser	1	€ 0,23	€ 0,23	correct, in kicad	U5	Connector_General	Package_DIP:DIP-8_W7.62mm_Socket		
GPS/GNSS	Module	GPS/GNSS	Quectel LC76GPAMD	Zuiniger, goedkoper bij groot aantal		half hole PCB breakout (1.1mm)	Mouser	1	€ 7,76	€ 7,76	correct, in kicad, niet op voorraad	U6	https://www.mouser.be/ProductDetail/Quectel/LC76GPAMD?qs=1	https://www.mouser.be/ProductDetail/Quectel/LC76GPAMD?qs=1		
GPS/GNSS	Module, alternatief	GPS/GNSS	Quectel LC76GPAMD	Zuiniger, goedkoper bij groot aantal		half hole PCB breakout (1.1mm)	Digikey	1	€ 8,50	€ 0,00	skip	U6	https://www.mouser.be/ProductDetail/Quectel/LC76GPAMD?qs=1	https://www.mouser.be/ProductDetail/Quectel/LC76GPAMD?qs=1		
GPS/GNSS	C, SMD	Ontkoppel (VCC)	CC0603KRX5R6BB106	VCC power circuit: zie hardware design p.26	10µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad, niet op voorraad	C29	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD	Ontkoppel (VCC)	CC0603KRX7R6BB104	VCC power circuit: zie hardware design p.26	100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C30	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD, alternatief	Ontkoppel (VCC)	CC0603KRX7R6BB104	VCC power circuit: zie hardware design p.26	100nF	0603	Mouser	1	€ 0,12	€ 0,12	correct, in kicad	C30	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD	Ontkoppel (VCC)	CC0603JRNPO9BN330	VCC power circuit: zie hardware design p.26	33pF	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	C31	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	TVS, SMD	ESD/surge bescherming VCC	SMBJ5.0CA-E3/52	VCC power circuit: zie hardware design p.26		SMB	Mouser	1	€ 0,49	€ 0,00	correct, skip, fout	-	Device:D_TVS	Diode_SMD:D_SMB		
GPS/GNSS	alternatief, SMD, TVS	ESD/surge bescherming VCC	SMAJ5.0CA-E3/61	kleiner Is bidirectioneel ok? Nee		SMA	Mouser	1	€ 0,41	€ 0,00	correct, fout, skip	D3	Device:D_TVS	Diode_SMD:D_SMA Diode_SMD:D_SMA_Handsoldering		
GPS/GNSS	C, SMD	Ontkoppel (V_BCKP)	CC0603KPX5R6BB475	V_BCKP power circuit: zie hardware design p.27 Multilayer Ceramic Capacitors MLCC - SMD/SMT 35V 4.7uF X5R 0603 10% HI CV	4,7µF	0603	Mouser	1	€ 0,24	€ 0,24	correct, in kicad	C33	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD, alternatief	Ontkoppel (V_BCKP)	CC0603KRX5R6BB475	V_BCKP power circuit: zie hardware design p.27 Multilayer Ceramic Capacitors MLCC - SMD/SMT 35V 4.7uF X5R 0603 10% HI CV	4,7µF	0603	Mouser	1	€ 0,20	€ 0,00	skip, correct, niet op voorraad, in kicad	C33	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD	Ontkoppel (V_BCKP)	CC0603KRX7R6BB104	V_BCKP power circuit: zie hardware design p.27	100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C34	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD, alternatief	Ontkoppel (V_BCKP)	CC0603KRX7R6BB104	V_BCKP power circuit: zie hardware design p.27	100nF	0603	Mouser	1	€ 0,12	€ 0,12	correct, in kicad	C34	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD	Ontkoppel (V_BCKP)	CC0603JRNPO9BN330	V_BCKP power circuit: zie hardware design p.27	33pF	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	C35	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering		
GPS/GNSS	C, SMD	V_BCKP backup (supercap)	CPH3225A-LF	GPS-backup: houdt RTC + ephemeris bij stroomuitval → warm start ~5 s i.p.v. ~30 s	0,1F	SMD (3225)	Mouser	1	€ 2,20	€ 2,20	correct, niet op voorraad, in kicad	C32	Device:C	Capacitor_SMD:C_1210_3225Metric_Pad1.33x2.70mm_HandSoldering		
GPS/GNSS	D, SMD	V_BCKP rectifier schottky	BAT54C	Dual Anode Common Cathode	30V 0,2A	SOT-23	Mouser	1	€ 0,10	€ 0,10	correct, in kicad	D2	Diode:BAT54C	Package_TO_SOT_SMD:SOT-23 Package_TO_SOT_SMD:SOT-23_Handsoldering		
GPS/GNSS	D, SMD	V_BCKP supercap -> 3,3V	BAT60AE6327HTSA1	Schottky Diode	Vf 300 mV	SOD-323-2	Mouser	1	€ 0,62	€ 0,62	correct, in kicad	D4	Device:D_Schottk	Diode_SMD:D_SOD-523		
GPS/GNSS	R, SMD	V_BCKP supercap -> 3,3V	RC0603FR-131KL	Om supercap op te laden	1K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R21	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering		
GPS/GNSS	R, SMD	Pullup (RESET)	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R20	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering		
GPS/GNSS	R, SMD	Pull down (D_SEL)	RC0603FR-1310KL	D_SEL low -> UART for communication	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R27	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering		
GPS/GNSS	TVS, SMD	ESD/surge bescherming V_BCKP	SMBJ5.0CA-E3/52	V_BCKP power circuit: zie hardware design p.27		SMB	Mouser	1	€ 0,49	€ 0,00	correct, skip, fout	-	Device:D_TVS	Diode_SMD:D_SMB		
GPS/GNSS	alternatief, TVS, SMD	ESD/surge bescherming V_BCKP	SMAJ5.0CA-E3/61	kleiner Is bidirectioneel ok? Nee		SMA	Mouser	1	€ 0,41	€ 0,00	correct, fout, skip	-	Device:D_TVS	Diode_SMD:D_SMA Diode_SMD:D_SMA_Handsoldering		

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Onder deel	Type	Beschrijving	Component	Opmerking	T _r Waarde	T _r Vorm	Distributeu r	#	📄 € (1)	📄 €€	🕒 Status	referenc e	Symbool	Footprint
GPS/GNSS	TVS, SMD	ESD/surge bescherming VCC en V_BCKP	SMF05CT1G	5 TVS dioden in één, 2 gebruikt	-	SC-88-6 (SOT-363-6)	Mouser	1	€ 0,38	€ 0,38	correct, in kicad	U8	https://www.snapeda.com/parts/SMF05CT1G/onsemi/view-part	https://www.snapeda.com/parts/SMF05CT1G/onsemi/view-part
GPS antenne	Antenne, Extern	FPC antenne	YFGA003AA	FPC antenne, RF 1 connector	-	FPC antenne (10cm kabel)	Digikey	1	€ 1,32	€ 1,32	correct, in kicad	-	-	-
GPS antenne	Connector, SMD	U.FL RF receptacle	U.FL-R-SMT-1(10)	U.FL/IPEX SMD receptacle; soldeer GPS FPC antennekabel	-	SMD	Mouser	1	€ 1,42	€ 1,42	correct, in kicad	AE1		Connector_Coaxial:U.FL_Hirose_U.FL-R-SMT-1_Vertical
GPS antenne	TVS, SMD	ESD protection device	ESD9B5.0ST5G	Lage junction-capacitance (<0,6pF) — zie hardware design p.40		SOD-923	Digikey	1	€ 0,19	€ 0,19	correct, in kicad	D3	Device:D_TVS	Diode_SMD:D_SOD-923
GPS antenne	C, SMD	Matching circuit	bepalen bij meting!	Pad voorzien, waarde bepaald na meting; zie hardware design p.40	DNP	0402		1		€ 0,00	uitzoeken, in kicad	C49	Device:C	Capacitor_SMD:C_0402_1005Metric Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSoldering
GPS antenne	C, SMD	Matching circuit	bepalen bij meting!	Pad voorzien, waarde bepaald na meting; zie hardware design p.40	DNP	0402		1		€ 0,00	uitzoeken, in kicad	C50	Device:C	Capacitor_SMD:C_0402_1005Metric Capacitor_SMD:C_0402_1005Metric_Pad0.74x0.62mm_HandSoldering
GPS antenne	R, SMD	Matching circuit	RC0402JR-070RL	zie hardware design p.40	0R	0402	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R28	Device:R	Resistor_SMD:R_0402_1005Metric Resistor_SMD:R_0402_1005Metric_Pad0.72x0.64mm_HandSoldering
E-ink	Module, Extern	1,54" 200x200 ePaper module	Waveshare 1.54" module	-		8-pin JST PH 2.0mm aansluiting	Waveshare	1		€ 0,00	correct, in kicad	J7	Zelf gemaakt	zie connector
E-ink	Connector	JST PH connector	8-pin JST PH 2.0mm	Meegeleverde kabel wordt geknipt en gekrimpt naar JST-PH uitgang	-	8-pin JST PH 2.0mm	-	1		€ 0,00	in bezit, in kicad	J7	zie module	Connector_JST:JST_PH_B8B-PH-K_1x08_P2.00mm_Vertical
Accelerometer	Module	Accelerometer	LIS2DW12 module	module bevat I²C pullups weerstanden (10K)	-	Module (2 4-pin headers)	Tinytronics	1		€ 0,00	correct, in kicad	H2	Zelf gemaakt	! zelf maken
Accelerometer	C, SMD	Ontkoppel	CC0603KPX5R6BB105	-	1µF	0603	Mouser	1	€ 0,11	€ 0,11	correct, in kicad	C23	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
Accelerometer	C, SMD	Ontkoppel	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C24	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
Accelerometer	C, SMD, alternatief	Ontkoppel	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,10	correct, in kicad	C24	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
Accelerometer	Header, Connector	4-pin header	2x 4-pin header		-	Twee 4-pin headers (2,54mm)	-	1		€ 0,00	in bezit, in kicad	H2	-	-
SD-kaart	Module	SD-kaart	SD adapter module	Module met level shifter en SD-houder, werkt met SPI	-	Module (8x2 pin header)	Tinytronics	1	€ 2,50	€ 2,50	correct	H3	Zelf gmaakt	PinHeader_2x08_P2.54mm_Horizontal PinHeader_2x08_P2.54mm_Vertical
SD-kaart	Header, Connector	SD-module connector	SFH11-PBPC-D08-RA-BK	Om verticaal te plaatsen Gebruiken??	-	Module (8x2 pin header)	Digikey	1	€ 1,13	€ 0,00	correct, skip	H3	-	PinHeader_2x08_P2.54mm_Horizontal PinHeader_2x08_P2.54mm_Vertical
SD-kaart	R, SMD	CS pullup	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R23	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
SD-kaart	R, SMD	MISO pullup	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R22	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
FRAM chip	IC, SMD	FRAM chip	FM24V01A-GTR	128-Kbit (16Kbyte)	-	8-SOIC	Digikey	1	€ 3,82	€ 3,82	correct, in kicad	IC4	https://www.mouser.be/ProductDetail/Infineon-Technologies/B24V01A-GTR of Memory_EEPR	https://www.mouser.be/ProductDetail/Infineon-Technologies/B24V01A-GTR Package_SO:SOIC-8_3.9x4.9mm_P1.27mm
FRAM chip	C, SMD	Ontkoppel (VDD)	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C36	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
FRAM chip	C, SMD, alternatief	Ontkoppel (VDD)	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,10	correct, in kicad	C36	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - li-ion lader	IC	BMS - li-ion lader	BQ25630YBGR	I²C 1-cell 5A buck battery charger with USB Type-C® detection	-	DSBGA (YBG)	Digikey	1	€ 3,37	€ 3,37	correct, in kicad, niet op voorraad	IC2	https://www.mouser.be/ProductDetail/Texas-Instruments/BQ25630YBGR	https://www.mouser.be/ProductDetail/Texas-Instruments/BQ25630YBGR
BMS - li-ion lader	IC, alternatief	BMS - li-ion lader	BQ25630YBGR	I²C 1-cell 5A buck battery charger with USB Type-C® detection		DSBGA (YBG)	Mouser	1	€ 3,45	€ 0,00	skip, correct, in kicad, niet op voorraad	IC2	https://www.mouser.be/ProductDetail/Texas-Instruments/BQ25630YBGR	https://www.mouser.be/ProductDetail/Texas-Instruments/BQ25630YBGR
BMS - li-ion lader	L, SMD	SW inductor	MAKK2520H1R0M	Datasheet: 1 µH typ, I_sat ≥ 2A	1µH	Shielded power, 2,5 x2,0mm	Mouser	1	€ 0,16	€ 0,16	correct, in kicad	L1	Device:L	Inductor_SMD:L_Taiyo-Yuden_NR-20xx (niet zeker) Inductor_SMD:L_Taiyo-Yuden_NR-20xx_HandSoldering (niet zeker) https://www.mouser.be/ProductDetail/TAIYO-YUDEN/MAKK2520H1R0M
BMS - li-ion lader	LED, SMD	Power good indicator	LTST-C191KGKT	Als een goede input source gedetecteerd is	groen	0603	Mouser	1	€ 0,13	€ 0,00	correct, skip	-	Device:LED	LED_SMD:LED_0603_1608Metric
BMS - li-ion lader	C, SMD	Ontkoppel (VBUS)	CC0603KPX5R6BB105	-	1µF	0603	Mouser	1	€ 0,11	€ 0,11	correct, in kicad	C1	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - li-ion lader	C, SMD	Ontkoppel (System load)	CC0603KRX5R6BB106	2 × 10µF parallel	2x 10µF	0603	Mouser	2	€ 0,22	€ 0,00	correct, fout, skip, niet op voorraad	-	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - li-ion lader	C, SMD, alternatief	Ontkoppel (System load)	CC0805MKX5R7BB226	22 µF op +5V_SERVO nabij servo connector; buffert stroompieken servo	22µF	0805	Mouser	1	€ 0,46	€ 0,46	correct, in kicad, niet op voorraad	C5	Device:C	Capacitor_SMD:C_0805_2012Metric Capacitor_SMD:C_0805_2012Metric_Pad1.18x1.45mm_HandSoldering
BMS - li-ion lader	C, SMD	Ontkoppel (PMID)	CC0603KRX5R6BB106	-	10µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad, niet op voorraad	C2	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering

nieuwe rij (bij selectie)		nieuwe rij ('alternatief')		randen tekenen		links >> distributeur		update prijzen (API)				update status 'skip'			
Onder deel	Type	Beschrijving	Component	Opmerking	T _r Waarde	T _r Vorm	Distributeu r	#	🔌 € (1)	🔌 €€	🔌 Status	referenc e	Symbool	Footprint	
BMS - li-ion lader	C, SMD	Ontkoppel (PMID)	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C3	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - li-ion lader	C, SMD, alternatief	Ontkoppel (PMID)	CC0603KRX7R6BB104		100nF	0603	Mouser	1	€ 0,12	€ 0,10	correct, in kicad	C3	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - li-ion lader	C, SMD	REGN - GND	CC0603KRX5R6BB475	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10V 4.7uF X5R 0603 10% HI CV	4,7µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad	C4	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - li-ion lader	C, SMD	SW - BTST	CC0603KRX7R7BB473	-	47nF	0603	Mouser	1	€ 0,10	€ 0,10	correct, in kicad	C7	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - li-ion lader	C, SMD	BAT - GND	CC0603KRX5R6BB106	-	10µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad, niet op voorraad	C6	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD	PG pullup	RC0603FR-072K2L		2,2K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R5	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD	INT pullup	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R6	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD	CE pulldown	RC0603FR-1310KL	-	10K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R29	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD	BATP - BAT	RC0603FR-07100RL		100R	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R7	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD	REGN - TS	RC0603FR-075K23L	JEITA-pullup met 103AT NTC (zie BQ25630 datasheet p.32)	5,23K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad, niet op voorraad	R8	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD	TS - GND	RC0603FR-0730K9L	Parallel met NTC voor JEITA-temperatuurschalen Berekend voor SMD NTC (NCP15XH103F03RC)	30,9K	0603	Mouser	1	€ 0,09	€ 0,00	correct, skip	R9	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	R, SMD, alternatief	TS - GND	RC0603FR-1330K1L	Parallel met NTC voor JEITA-temperatuurschalen Berekend voor 103AT-2 (zie datasheet p.32)	30,1K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R9	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - li-ion lader	NTC, SMD	Batterij temp sensor (TS - GND)	NCP15XH103F03RC	103AT-compatibel — in battery pack of nabij batterijconnector	10K @25°C, β=3380	0402	Mouser	1	€ 0,09	€ 0,00	correct, skip, niet op voorraad	TH1	Device:Thermisto	Resistor_SMD:R_0402_1005Metric	
BMS - li-ion lader	NTC, THT, alternatief	Batterij temp sensor (TS - GND)	103AT-2	THT, duurder, makkelijker tegen batterij te plaatsen	10K @25°C, β=3435 K	Radial, 2,54mm spacing	Mouser	1	€ 0,58	€ 0,58	correct, in kicad	TH1	Device:Thermisto	Capacitor_THT:C_Disc_D3.4mm_W2.1mm_P2.50mm	
BMS - li-ion lader	Knop, THT	System power reset control (QON - GND)	TS02-66-43-BK-160-LCR-	optioneel	-	SPST momentary tactile, 6x6 mm THT		1		€ 0,00	skip	-	Switch:SW_Push	Button_Switch_THT:SW_PUSH_6mm	
BMS - 3,3V buck-boost	IC	3,3V buck-boost converter	RT6158AWSC	5µA ruststroom		DFN-10	Mouser	1	€ 1,66	€ 1,66	correct, in kicad	U1	https://app.ultralib.com/details/d01e2d83-aa6c-11ec-903d-000000000000	https://app.ultralibrarian.com/details/d01e2d83-aa6c-11ec-903d-000000000000	
BMS - 3,3V buck-boost	L, SMD	Power inductor	MAKK2520H1R0M	Datasheet van buck-boost: 1 µH typ	1µH	Shielded power, 2,5 x2,0mm	Mouser	1	€ 0,16	€ 0,16	correct, in kicad	L2	Device:L	Inductor_SMD:L_Taiyo-Yuden_NR-20xx (niet zeker) Inductor_SMD:L_Taiyo-Yuden_NR-20xx_HandSoldering (niet zeker) https://www.mouser.be/ProductDetail/TAIYO-YUDEN/MAKK2520H1R0M	
BMS - 3,3V buck-boost	C, SMD	Ontkoppel Cin	CC0603KRX5R6BB106	2x 10µF	10µF	0603	Mouser	2	€ 0,22	€ 0,45	correct, in kicad, niet op voorraad	C44, C45	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - 3,3V buck-boost	C, SMD	Ontkoppel Cout	CC0603KRX5R6BB106	4x 10µF	10µF	0603	Mouser	4	€ 0,22	€ 0,90	correct, in kicad, niet op voorraad	C9, C46, C47, C48	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - 3,3V buck-boost	C, SMD	CFF (Vout - FB)	CC0603JRNPO9BN560		56pF	0603	Mouser	1	€ 0,10	€ 0,10	correct, in kicad	C8	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering	
BMS - 3,3V buck-boost	R, SMD	R1 (Vout - FB)	RC0603FR-071ML		1M	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R10	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - 3,3V buck-boost	R, SMD	R2 (FB - GND) - 1% tol	RC0603FR-07294KL	1% tolerantie aangeraden voor Vout-precisie	294K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R11	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - 3,3V buck-boost	R, SMD	Ren (EN - GND)	RC0603FR-07100KL	Skip: altijd enabled -> verbonden met +SYS	100K	0603	Mouser	1	€ 0,09	€ 0,00	correct, skip	-	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering	
BMS - 3,3V LDO regulator	IC	3,3V LDO regulator	TCR3UG33A,LF	Met auto discharge	-	WCSP4F	Mouser	1	€ 0,27	€ 0,27	correct, in kicad	U4	https://www.snapeda.com/parts/TCR3UG33A%2CLF/Toshiba%20Semiconductors	https://www.snapeda.com/parts/TCR3UG33A%2CLF/Toshiba%20Semiconductors	
BMS - 3,3V LDO regulator	C, SMD	Ontkoppel (Vin)	CC0603KPX5R6BB105	-	1µF	0603	Mouser	1	€ 0,11	€ 0,11	correct, in kicad	C11	Device:C	Capacitor_SMD:C_0603_1608Metric	
BMS - 3,3V LDO regulator	C, SMD	Ontkoppel (Vout)	CC0603KPX5R6BB105	-	1µF	0603	Mouser	1	€ 0,11	€ 0,11	correct, in kicad	C10	Device:C	Capacitor_SMD:C_0603_1608Metric	
BMS - 5V boost	IC	5V boost converter	TPS61253F	4,5A spoelstroom	-	DSBGA-9	Digikey	1	€ 1,00	€ 1,00	correct, in kicad	U4	Zelf gemaakt	https://www.snapeda.com/parts/DRV2625YFFR/Texas%20Instruments%20Semiconductors ander component, zelfde package	
BMS - 5V boost	IC, alternatief	5V boost converter	TPS61253F	4,5A spoelstroom	-	DSBGA-9	Mouser	1	€ 0,44	€ 0,00	skip, niet op voorraad	U4	Zelf gemaakt	https://www.snapeda.com/parts/DRV2625YFFR/Texas%20Instruments%20Semiconductors ander component, zelfde package	

nieuwe rij (bij selectie)		nieuwe rij ('alternatief')		randen tekenen	links >> distributeur		update prijzen (API)				update status 'skip'			
Onder deel	Type	Beschrijving	Component	Opmerking	T _r Waarde	T _r Vorm	Distributeur	#	📄 € (1)	📄 €€	🕒 Status	referentie	Symbool	Footprint
BMS - 5V boost	L, SMD	Power inductor	MAKK2520H1R0M	Datasheet p.17: 1 µH typ Max 3A ok??	1µH	Shielded power, 2,5 x2,0mm	Mouser	1	€ 0,16	€ 0,16	correct, in kicad	L3	Device:L	Inductor_SMD:L_Taiyo-Yuden_NR-20xx (niet zeker) Inductor_SMD:L_Taiyo-Yuden_NR-20xx_HandSoldering (niet zeker) https://www.mouser.be/ProductDetail/TAIYO-YUDEN/MAKK2520H1R0M
BMS - 5V boost	C, SMD	Ontkoppel Cin	CC0603KRX5R6BB475	Datasheet p.17: 1 µH typ Multilayer Ceramic Capacitors MLCC - SMD/SMT 10V 4.7uF X5R 0603 10% HI CV	4,7µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad	C12	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - 5V boost	C, SMD	Ontkoppel Cout1	CC0603KRX5R6BB106	Datasheet p.17: 1 µH typ	10µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad, niet op voorraad	C13	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - 5V boost	C, SMD	Ontkoppel Cout2	CC0603KRX5R6BB475	Datasheet p.17: 1 µH typ Multilayer Ceramic Capacitors MLCC - SMD/SMT 10V 4.7uF X5R 0603 10% HI CV	4,7µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad	C42	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - 5V boost	C, SMD	Ontkoppel Cout3	CC0603KRX5R6BB475	Datasheet p.17: 1 µH typ Multilayer Ceramic Capacitors MLCC - SMD/SMT 10V 4.7uF X5R 0603 10% HI CV	4,7µF	0603	Mouser	1	€ 0,22	€ 0,22	correct, in kicad	C43	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
BMS - 5V boost	R, SMD	Pull down (EN)	RC0603FR-07100KL		100K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R12	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
BMS - 5V boost	R, SMD, alternatief	Pull up (MODE)	RC0603FR-07100KL	MODE->5V = forced PWM mode	100K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R13	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
BMS - USB-C	Connector, SMD	USB-C connector	USB4105-GF-A-060	USB 2.0 Type-C 16-pin top-mount; voedingsingang van de lader	-	SMD	Digikey	1	€ 0,70	€ 0,00	correct, skip	J1	Connector_USB:L	Connector_USB:GCT_USB4105-GF-A_TopMnt_Horizontal
BMS - USB-C	Connector, SMD, alternatief	USB-C connector	USB4105-GF-A-060	USB 2.0 Type-C 16-pin top-mount; voedingsingang van de lader	-	SMD	Mouser	1	€ 0,80	€ 0,80	correct, in kicad	J1	Connector_USB:L	Connector_USB:GCT_USB4105-GF-A_TopMnt_Horizontal Connector_USB:USB_C_Receptacle_GCT_USB4105-xx-A_16P_TopMount
BMS - USB-C	Connector, SMD, alternatief	USB-C connector	UJ20-C-H-G-MSMT-3B-P16-TR	USB 2.0 Type-C 16-pin top-mount; voedingsingang van de lader Vervangen indien op voorraad??	-	SMD	Mouser	1	€ 0,27	€ 0,00	nog nazien, niet op voorraad, skip	-	Connector_USB:L	
BMS - USB-C	TVS, SMD	USB ESD bescherming	USBLC6-2SC6	Plaatsen direct naast USB-C connector; beschermt D+/D- lijnen	-	SOT-23-6	Digikey	1	€ 0,16	€ 0,16	correct, in kicad	U7	Device:D_TVS	Package_TO_SOT_SMD:SOT-23-6 Package_TO_SOT_SMD:SOT-23-6_Handsoldering
BMS - USB-C	TVS, SMD, alternatief	USB ESD bescherming	USBLC6-2SC6	Plaatsen direct naast USB-C connector; beschermt D+/D- lijnen	-	SOT-23-6	Mouser	1	€ 0,42	€ 0,00	correct, skip	U7	Device:D_TVS	Package_TO_SOT_SMD:SOT-23-6 Package_TO_SOT_SMD:SOT-23-6_Handsoldering
BMS - USB-C	R, SMD	CC1 + CC2 pull-down	RC0603FR-135K1L	2x 5,1 kΩ naar GND; vereist voor USB-C brondetectie door BQ25630 (zonder: max 100 mA) Fout! Dat zit al verwerkt in de bq25630 lader	5,1K	0603	Mouser	2	€ 0,09	€ 0,00	correct, fout, skip	R1, R2	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
BMS - Batterij	Connector, THT	Batterijconnector	S2B-PH-K-S(LF)(SN)	JST PH 2.0mm 2-pin rechte THT; overeenkomt met standaard Li-ion kabel	-	JST PH 2.0mm 2-pin	Digikey	1	€ 0,10	€ 0,00	skip, in bezit	BT1	Connector_JST:JS	Connector_JST:JST_PH_S2B-PH-K_1x02_P2.00mm_Horizontal
Motor	Motor, Extern	Servo motor	TD-8120MG Digital Servo	waterproof - 20kg	-	Zie datasheet	Tinytronics	1	€ 10,25	€ 10,25	correct, in kicad	H4	Zelf gemaakt	Zie connector
Motor	Connector	Servo motor	JST-PH 1x03 2mm	Kabel knippen en JST-PH op krimpen	-	JST-PH 1x03 2mm	-	1	€ 0,00	€ 0,00	in bezit, in kicad	H4	Zie motor	Connector_JST:JST_PH_B3B-PH-K_1x03_P2.00mm_Vertical
Motor	R, SMD	Current sense	PT1206FR-7W0R1L	Stall-stroom servo ≈ 1A; bevestig waarde a.h.v. max stroom × ADC-range	0,1R / 500mW	1206	Mouser	1	€ 0,32	€ 0,32	correct, in kicad	R16	Device:R	Resistor_SMD:R_1206_3216Metric Resistor_SMD:R_1206_3216Metric_Pad1.30x1.75mm_HandSoldering
Motor	R, SMD	Current sense RC filter	RC0603FR-071KL	RC filter met 4,8kHz cutoff	1K	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R26	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
Motor	C, SMD	Current sense RC filter	CC0603KRX7R6BB333	idem	33nF	0603	Mouser	1	€ 0,10	€ 0,10	correct, in kicad	C22	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
Motor	C, SMD	Servo rail bulk (groot)	CC0805MKX5R7BB226	22 µF op +5V_SERVO nabij servo connector; buffert stroompieken servo	22µF	0805	Mouser	1	€ 0,46	€ 0,00	correct, fout, skip, niet op voorraad	-	Device:C	Capacitor_SMD:C_0805_2012Metric
Motor	C, THT, alternatief	Servo rail bulk (groot)	UVR1C471MPD1TD	470 µF is beter voor rail sag te voorkomen	470µF	8 x 11,5 radial, lead spacing 3.5 mm	Mouser	1	€ 0,34	€ 0,34	correct, in kicad	C21	Device:C	Capacitor_THT:CP_Radial_D8.0mm_P3.50mm
Motor	C, SMD	Servo rail bulk (klein)	CC0603KRX7R6BB104	100 nF op +5V_SERVO nabij servo connector	100nF	0603	Mouser	1	€ 0,12	€ 0,00	correct, in kicad, skip	C20	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
Motor	C, SMD, alternatief	Servo rail bulk (klein)	CC0603KRX7R6BB104	100 nF op +5V_SERVO nabij servo connector	100nF	0603	Mouser	1	€ 0,12	€ 0,12	correct, in kicad	C20	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering
LED	LED	Kleur-LED (rood, groen)	BL-BEG204-7-E	Rood en groen, difuus, THT, common anode	-	5mm LED, radiaal, 3 pinnen	Digikey	1	€ 0,17	€ 0,17	correct, in kicad	D1	LED_DUAL_KAK	LED_D5.0mm-3_Horizontal_O3.81mm_Z3.0mm
LED	R, SMD	Stroomlimiet (rood)	RC0603FR-07150RL		150R	0603	Mouser	1	€ 0,09	€ 0,00	correct, fout, skip	R14	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
LED	R, SMD	Stroomlimiet (rood)	AF0603FR-07160RL		160R	0603	Mouser	1	€ 0,09	€ 0,09	correct, in kicad	R14	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
LED	R, SMD	Stroomlimiet (groen)	RC0603FR-07130RL		130R	0603	Mouser	1	€ 0,10	€ 0,10	correct, in kicad	R15	Device:R	Resistor_SMD:R_0603_1608Metric Resistor_SMD:R_0603_1608Metric_Pad0.98x0.95mm_HandSoldering
Hall effect sensor	Hall sensor	Hall effect sensor	DRV5032DULPGM	Unipolair: Z-pool, THT 2 dezelfde voor prototype, noord en zuid voor finaal ontwerp	-	TO-92	Mouser	2	€ 0,30	€ 0,60	correct, in kicad	IC1, IC3	https://www.mou	https://www.mouser.be/ProductDetail/Texas-Instruments/DRV5032DULPGM
Hall effect sensor	C, SMD	Ontkoppel (VCC)	CC0603KRX7R6BB104	één per sensor	100nF	0603	Mouser	2	€ 0,12	€ 0,00	correct, in kicad, skip	C25, C26	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandSoldering

nieuwe rij (bij selectie)		nieuwe rij ('alternatief')		randen tekenen	links >> distributeur		update prijzen (API)				update status 'skip'			
Onder deel	 Type	Beschrijving	Component	Opmerking	 Waarde	 Vorm	Distributeur	#	 € (1)	 € €	 Status	referentie	Symbool	Footprint
Hall effect sensor	C, SMD, alternatief	Ontkoppel (VCC)	CC0603KRX7R6BB104	één per sensor	100nF	0603	Mouser	1	€ 0,12	€ 0,12	correct, in kicad	C25, C26	Device:C	Capacitor_SMD:C_0603_1608Metric Capacitor_SMD:C_0603_1608Metric_Pad1.08x0.95mm_HandS
ST-link	Module, Extern	ST-link	STLINK-V3MINIE	Programmer/debugger	-	Standard Card Edge Connectors Dual Row Inverted Card Edge 2x5pos w/ Kapton	Mouser	1	€ 22,39	€ 0,00	correct, in kicad, skip	J3	https://www.mouser.com/ProductDetail/KYOCERA-AVX/0091590	Zie connector
ST-link	Module, Extern, alternatief	ST-link	STLINK-V3MINIE	Programmer/debugger	-	Standard Card Edge Connectors Dual Row Inverted Card Edge 2x5pos w/ Kapton	Farnell	1	€ 28,47	€ 28,47	correct, in kicad	J3	https://www.mouser.com/ProductDetail/KYOCERA-AVX/0091590	Zie connector
ST-link	Connector	ST-link	Card edge connector	Programmer/debugger	-	Standard Card Edge Connectors Dual Row Inverted Card Edge 2x5pos w/ Kapton	Mouser	1	€ 1,21	€ 1,21	correct, in kicad	J3	Zie ST-link	https://www.mouser.be/ProductDetail/KYOCERA-AVX/0091590
Totaal									27/08/2026	€ 120,46				